

## SCREENING TEST - JUNIOR LEVEL - 1997

1. A man sells a watch and a radio for Rs.990 each. On the radio he gains 10% and on the watch he loses 10%. His net gain or loss is  
(A) 1% loss (B) 1% gain (C) No loss or gain (D) 2% gain
2. A cube of size  $5 \times 5 \times 5$  is painted blue on all its six sides. The cube is cut uniformly into 125 cubes of size  $1 \times 1 \times 1$ . The number of cubes with only one face coloured is  
(A) 40 (B) 36 (C) 45 (D) 54
3. If the side of a cube is increased by 10%, volume of the cube increases by  
(A) 30% (B) 10% (C) 22.1% (D) 33.1%
4. If  $\frac{4 + 3\sqrt{3}}{\sqrt{7 + 4\sqrt{3}}} = a + \sqrt{b}$  where  $a, b$  are integers, then  $a, b$ , respectively, are  
(A) 12, 1 (B) 1, 12 (C) -1, 12 (D) -12, 1
5. The value of  $k$  for which  $(x + 2)$  is a factor of  $(x + 1)^7 + (2x + k)^3$  is  
(A) -5 (B) 5 (C) -1 (D)  $-4 - 3^{\frac{7}{3}}$
6. The value of  $\frac{12}{3 + \sqrt{5} - 2\sqrt{2}}$  is  
(A)  $1 + \sqrt{5} + \sqrt{10} + \sqrt{2}$  (B)  $1 + \sqrt{5} - \sqrt{10} + \sqrt{2}$   
(C)  $1 + \sqrt{5} + \sqrt{10} - \sqrt{2}$  (D)  $1 - \sqrt{5} - \sqrt{10} + \sqrt{2}$
7. The greatest of the numbers  $\sqrt{2}$ ,  $\sqrt[3]{9}$ ,  $\sqrt[4]{11}$ ,  $\sqrt[6]{17}$  is  
(A)  $\sqrt{2}$  (B)  $\sqrt[3]{9}$  (C)  $\sqrt[4]{11}$  (D)  $\sqrt[6]{17}$
8. If  $x = 7 + 4\sqrt{3}$ , and  $xy = 1$ , then  $\frac{1}{x^2} + \frac{1}{y^2}$  is  
(A) 64 (B) 134 (C) 194 (D) 234
9. The sum of numbers  $n$  such that  $100 \leq n \leq 500$  and  $n$  is not divisible by 6 is

10. The smallest positive number with exactly 10 divisors is (for a number  $n$ , 1 and  $n$  are also counted as divisors)
- (A) 512 (B) 162 (C)  $3^9$  (D) none of these
11. If different letters represent different digits and  $OX \times OX = FOX$ ,  $X$  represents
- (A) 5 (B) 1 or 6 (C) 1 or 5 (D) 5 or 6
12. The length of one of the legs of a right angled triangle exceeds the length of the other leg by 10 cm but is smaller than that of the hypotenuse by 10 cm. The length of the hypotenuse of the triangle in cms is
- (A) 20 (B) 30 (C) 40 (D) 50
13. If  $a_1, a_2, \dots, a_{12}$  are positive real numbers such that  $\frac{a_1}{a_2} = \frac{a_2}{a_3} = \dots = \frac{a_{11}}{a_{12}}$ ,  $a_1 + a_2 + a_3 + a_4 = 30$  and  $a_5 + a_6 + a_7 + a_8 = 480$ , then  $a_9 + a_{10} + a_{11} + a_{12}$  is
- (A) 19680 (B) 96180 (C) 7680
- (D) can not be determined from the given information.
14. The number of germs in a test tube doubles every four minutes. If at the end of 40 minutes, there were 1024 germs, at the end of how many minutes were there exactly 256 germs?
- (A) 32 min. (B) 24 min. (C) 12 min. (D) 8 min.
15. Circles of radii 1 cm. are described with centers at the four corners  $A, B, C, D$  of a square of side 2 cm. The radius in cm. of the circle which touches all these four circles externally is
- (A)  $\sqrt{2} - 1$  (B)  $\sqrt{2} + 1$  (C)  $\frac{1}{\sqrt{2}} + 1$  (D)  $1 - \frac{1}{\sqrt{2}}$
16. In a triangle  $ABC$ ,  $BE, CF$  are medians.  $BE = 9$  cm and  $CF = 12$  cm. If  $BE$  is perpendicular to  $CF$ , the area of the triangle  $ABC$  in sq.cms is
- (A) 36 (B) 72 (C) 24

17. The largest number that leaves the same remainder while dividing 4813, 4429, 4957 is

- (A) 12 (B) 48 (C) 96 (D) 192

18. If  $\alpha, \beta$  are roots of  $(x - \sqrt{2})(x - \sqrt{3}) = \sqrt{5}$ , the roots of  $(x - \alpha)(x - \beta) + \sqrt{5} = 0$  are

- (A)  $\sqrt{2}, \sqrt{5}$  (B)  $\sqrt{3}, \sqrt{5}$  (C)  $\sqrt{2}, \sqrt{3}$  (D)  $\sqrt{2} + \sqrt{3}, \sqrt{3} + \sqrt{5}$

19. A regular polygon has 44 diagonals. The number of sides is

- (A) 9 (B) 10 (C) 11 (D) 12

20. The sum of first 2000 terms of the series

$$\frac{1}{2} + \frac{3}{4} + \frac{7}{8} + \frac{15}{16} + \dots \text{ is}$$

- (A)  $2^{2000} - 2001$  (B)  $1 - 2^{-2000}$  (C)  $1999 + 2^{-2000}$  (D)  $2^{2000} - 1$

21. If  $\sqrt{7x} - \sqrt{3x} = 4$ ,  $x$  equals

- (A)  $10 + 2\sqrt{21}$  (B)  $10 - 2\sqrt{21}$  (C)  $\sqrt{7} + \sqrt{3}$  (D)  $\sqrt{7} - \sqrt{3}$

22. For real numbers  $x, y$  define  $x * y = xy - x + y$ . If  $5 * x = 17$ ,  $x$  equals

- (A)  $3\frac{2}{5}$  (B) 2 (C) 3 (D)  $3\frac{2}{3}$

23. The smaller angle between the hands of a clock at 12:35 is

- (A)  $167\frac{1}{2}^\circ$  (B)  $150^\circ$  (C)  $165^\circ$  (D)  $175^\circ$

24. The least number of colours needed to paint the six faces of a cube so that no two adjacent faces (that is faces that have an edge in common) have the same colour is

- (A) 2 (B) 6 (C) 4 (D) 3

25. Each of the six faces of a cube is numbered with a different integer. Each vertex is then assigned a vertex number which is the sum of the numbers on the faces that meet at that vertex. The sum of all the vertex numbers is



26. All the corners of a cube of volume 1000 cubic centimeters are cut off neatly creating a new triangular face at each vertex. Each of these triangular faces is an equilateral triangle with side 1 cm. The number of edges on the new solid is

- (A) 24 (B) 12 (C) 16 (D) 36

27. The set of perfect squares  $\{1^2, 2^2, 3^2, \dots\}$  is divided into groups as follows:  $G_1 = \{1^2\}$ ,  $G_2 = \{2^2, 3^2, 4^2\}$ ,  $G_3 = \{5^2, 6^2, 7^2, 8^2, 9^2\}$ , and so on. The first number in  $G_{100}$  is

- (A)  $(9801)^2$  (B)  $(9800)^2$  (C)  $(9802)^2$  (D)  $(10000)^2$

28. The wheels of a truck travelling at 60 kmph. make four revolutions per second. The diameter of each wheel in meters is

- (A)  $\frac{25}{6\pi}$  (B)  $\frac{100}{6\pi}$  (C)  $\frac{25\pi}{6}$  (D)  $\frac{25}{12\pi}$

29.  $\underbrace{(66 \dots 6)}_n^2 + \underbrace{88 \dots 8}_n$  equals

- (A)  $\underbrace{44 \dots 4}_{2n}$  (B)  $\underbrace{44 \dots 4}_{2n-2}$  (C)  $\underbrace{44 \dots 4}_{2n-1}$  (D)  $\underbrace{2424 \dots 24}_{2n}$

30. The number of integers  $x$  for which  $x^2 + 6x + 12$  is a perfect square is

- (A) 0 (B) 1 (C) 2 (D) 3